

Exchange Rate Movements and the Australian Economy:

A Replication of Manalo et al. (2015)

Branesh Prakash

Monash University

Student ID: 35775246

Summary

This paper replicates the aggregate structural vector autoregression (SVAR) model of Manalo et al. (*Economic Modelling* 2015; 47: 53-62). The narrow replication successfully characterises the paper’s transmission effects of exchange rate movements in the Australian economy and reproduces findings that exchange rate shocks explain little of Australian macroeconomic volatility and foreign shocks dominate Australia’s long-run real exchange rate variation, although some quantitative differences exist owing to data differences. The wider replication primarily extends the sample and reveals that the transmission mechanisms remain broadly the same, however, the decomposition of exchange-rate movements changes more materially, whereby foreign factors account for a smaller share of medium- and long-run real exchange rate variation in extended samples. Controlling for pandemic-related output contractions proved important for recovering the pre-pandemic output response in the post-2019 period.

1 Introduction

Exchange rate movements are an important source of adjustment in small open economies such as Australia. By changing the relative price of domestically and foreign-produced goods and services, they can affect economic activity, inflation and monetary policy. Importantly, however, the effects of an exchange rate movement depend on what caused it. An appreciation associated with stronger commodity prices or foreign demand need not have the same consequences as an appreciation that is unrelated to these economic fundamentals. Manalo et al. (2015) examine this distinction using a structural vector autoregression (SVAR) of the Australian economy. Their results suggest that a temporary 10 percent exogenous appreciation of the real exchange rate lowers output and inflation and induces an easing of monetary policy. They also find that movements in Australia’s real exchange rate are predominantly driven by foreign economic developments, supporting the interpretation of the floating exchange rate as a shock absorber rather than an important independent source of macroeconomic volatility.

This paper undertakes both a narrow and a wider replication of the aggregate results in Manalo et al. (2015). The narrow replication reproduces their six-variable SVAR over the original sample, retaining the small-open-economy restrictions, recursive identification and lag specification used in the paper. The central qualitative findings are reproduced: an exogenous appreciation lowers Australian real GDP and inflation and is followed by a decline in the cash rate, while foreign shocks become increasingly important in explaining exchange-rate variation at longer horizons. The magnitudes are not identical to those reported by Manalo et al., particularly for the inflation and monetary policy responses, but the direction and broad dynamics of the responses are similar.

The wider replication examines whether these conclusions remain robust when the model is estimated using more recent data. Extending the sample up to the eve of the COVID-19 pandemic produces results that remain close to the original transmission mechanism. Extending the model mechanically through the pandemic, however, substantially changes the estimated output response. This appears to reflect the

unusually large macroeconomic disturbances associated with COVID-19 rather than a straightforward change in the underlying exchange-rate transmission mechanism. Once we control for certain pandemic observations, the contractionary response of output to an appreciation re-emerges, although some aspects of the inflation and monetary policy responses differ from those obtained over the original sample.

As an additional extension and similar to Manalo et al. (2015), the foreign block of the SVAR is augmented with the CBOE Volatility Index (VIX) to account for global financial risk explicitly and assess whether the baseline exchange-rate shock partly captures changes in global financial risk. Including the VIX leaves the main qualitative responses of domestic activity and inflation to an exchange-rate shock broadly intact, but changes the attribution of short-run exchange-rate fluctuations.

The rest of this paper is organised as follows. Section 2 provides the model specification, Section 3 outlines the data and replication design, Section 4 presents results and Section 5 concludes the outcome of the replication exercise.

2 Model Specification

Following Manalo et al. (2015), we estimate a structural vector autoregression (SVAR) for the Australian economy. Let

$$Y_t = [F_t' \quad D_t']'$$

where F_t contains the foreign variables and D_t contains the Australian variables. For the aggregate model,

$$F_t = [y_t^{US} \quad tot_t]', \quad D_t = [y_t^{AU} \quad \pi_t \quad i_t \quad q_t]'$$

where y_t^{US} is US real GDP, tot_t is Australia's terms of trade, y_t^{AU} is Australian real GDP, π_t is trimmed mean inflation, i_t is the cash rate, and q_t is the real trade-weighted index of the Australian dollar (i.e., the AUD RTWI). An increase in q_t denotes an appreciation of the Australian dollar.

The structural VAR can be written as

$$A_0 Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \varepsilon_t,$$

where A_0 captures contemporaneous relationships between the variables and ε_t is a vector of mutually orthogonal structural shocks. For convenience, the structural shocks are normalised such that $E(\varepsilon_t \varepsilon_t') = I$. The corresponding reduced-form VAR is

$$Y_t = \mu + \Phi_1 Y_{t-1} + \Phi_2 Y_{t-2} + e_t,$$

where e_t is the vector of reduced-form innovations with $E(e_t e_t') = \Sigma$.

The reduced-form innovations are related to the structural shocks according to

$$e_t = B \varepsilon_t,$$

where $B = A_0^{-1}$ is the contemporaneous impact matrix. It follows from the normalisation of the structural shocks that $BB' = \Sigma$.

The reduced-form VAR can be estimated directly, since only lagged endogenous variables appear on its right-hand side. However, the reduced-form residuals may be contemporaneously correlated and therefore

cannot themselves be given a causal interpretation. Identification requires restrictions that recover the structural shocks from these reduced-form innovations.

The model treats Australia as a small open economy. Accordingly, developments in the Australian economy are assumed not to affect the foreign block, either contemporaneously or with a lag. In reduced-form notation, the lag structure can therefore be represented as

$$\begin{bmatrix} F_t \\ D_t \end{bmatrix} = c + \sum_{j=1}^2 \begin{bmatrix} \Phi_{FF,j} & 0 \\ \Phi_{DF,j} & \Phi_{DD,j} \end{bmatrix} \begin{bmatrix} F_{t-j} \\ D_{t-j} \end{bmatrix} + \begin{bmatrix} e_t^F \\ e_t^D \end{bmatrix}.$$

The zero in the upper-right block imposes the small-open-economy restriction: Australian variables may respond to foreign developments, but Australian developments do not feed back into US output or Australia's terms of trade. The terms of trade are treated as foreign because Australia is assumed to be a price-taker in its major export markets. This restriction allows foreign shocks to propagate to the Australian economy while ruling out feedback from the domestic block to the foreign block. The foreign variables are important for the identification of the exchange-rate shock. US output captures external economic conditions, while the terms of trade capture an important source of variation in the Australian dollar associated with commodity prices. Controlling for these factors reduces the risk of interpreting exchange-rate movements driven by external fundamentals as autonomous exchange-rate shocks.

The structural shocks are identified recursively using a Cholesky decomposition of the reduced-form covariance matrix Σ . The baseline ordering is

$$y_t^{US} \rightarrow tot_t \rightarrow y_t^{AU} \rightarrow \pi_t \rightarrow i_t \rightarrow q_t.$$

Equivalently, the impact matrix B is lower triangular. Of particular interest is the exchange-rate shock, which is identified by ordering the RTWI, q_t , last. This allows the exchange rate to respond immediately to innovations in all other variables in the system, consistent with the Australian dollar being determined in a liquid financial market that can rapidly incorporate new macroeconomic information. In contrast, an exchange-rate shock is assumed not to affect output, inflation or the cash rate within the same quarter; its effects on these variables occur with a lag. The identified exchange-rate shock can therefore be interpreted as the component of a movement in the real exchange rate that cannot be explained by contemporaneous or lagged movements in the other macroeconomic variables included in the model.

Additionally, the ordering of the cash rate before the exchange rate allows the exchange rate to react immediately to monetary policy innovations, while ruling out a contemporaneous direct policy-rate response to an exchange-rate shock. Monetary policy may nevertheless respond subsequently as the exchange-rate movement affects output and inflation. Manalo et al. (2015) emphasised that it takes time for firms to alter pricing and production decisions and existing literature finds that exchange rate movements do not have a large direct influence on Australian monetary policy decisions.

Given the recursive identification, the VAR has the vector moving-average representation $Y_t = \Psi_0 \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + \dots$, where Ψ_h describes the response of the variables h quarters after a structural shock. The impulse responses reported below are scaled so that the exchange-rate shock produces a 10 percent appreciation of the RTWI. Forecast error variance decompositions are also used to assess both the importance of exchange-rate shocks for domestic macroeconomic fluctuations and the relative importance of foreign, domestic and exchange-rate shocks in explaining movements in the RTWI.

3 Data and replication design

3.1 Data and transformations

The analysis uses quarterly data, whereby the narrow replication is estimated over the original sample, 1985Q1–2013Q2, while the dataset assembled for the wider replication extends through 2026Q2. US real GDP is obtained from the Federal Reserve Economic Data (FRED) database, Australia’s terms of trade and real GDP are sourced from the Australian Bureau of Statistics (ABS), whilst Australia’s trimmed mean inflation, cash rate and real trade-weighted index are obtained from the Reserve Bank of Australia (RBA). The VIX, used in the additional global-risk specification, is obtained from the Chicago Board Options Exchange (CBOE) and is available from 1990Q1.

The construction of the variables follows Manalo et al. (2015) as closely as possible. US and Australian real GDP, the terms of trade and the RTWI are transformed as 100 times their natural logarithms. US and Australian GDP are then expressed as deviations from their respective quadratic time trends estimated over the relevant sample. The terms of trade and RTWI enter the VAR in log levels. Trimmed mean inflation enters as per its original seasonally adjusted quarterly percentage change from the data source and similarly, the cash rate enters in its original rate levels. Although quarterly inflation is estimated in the VAR, its impulse response is converted to year-ended inflation by summing the responses over four quarters, following the original paper.

The quarterly cash-rate series required some additional construction because a single consistent series is not available over the entire sample. From 1985Q1 to 1990Q3, the RBA’s official-market 11 am call rate is used, averaging the underlying observations to quarterly frequency. From 1990Q4 to 1998Q2 the quarterly average of the cash rate target is then used, and from 1998Q3 onwards, the quarterly average of the interbank overnight cash rate is taken. This construction is intended to reproduce as closely as possible the policy-rate measure specified in and used by Manalo et al. (2015), and appears consistent with how it is described in other papers (Poke and Wells 2007).

The baseline VAR uses the deterministic controls as per the original paper. These include an inflation-targeting dummy, which takes the value one from 1993Q1 onwards reflecting structural break related to start of inflation targeting, and separate dummies for 2008Q4, 2009Q1, 2009Q2 and 2009Q3 to control for the unusually volatile period surrounding the global financial crisis. Two lags of the endogenous variables are included. The same transformations are applied in the extended specifications, with the quadratic GDP trends re-estimated over each estimation sample.

The replication therefore uses currently available FRED, ABS and RBA series rather than the data available to the original authors when analysis was done. Consequently, revised historical observations, together with differences in the construction of the early cash-rate series, may contribute to some numerical differences between the replicated and published results.

3.2 Replication design

The narrow replication retains the original six-variable aggregate SVAR, sample period, lag structure, deterministic controls and recursive identification, and focuses on reproducing the aggregate exchange-rate impulse responses and forecast error variance decompositions. The wider replication first extends the sample endpoint to 2019Q4, immediately before the COVID-19 pandemic, and then to 2026Q2. As the full sample contains unusually large pandemic-era disturbances, we also estimate a specification controlling for the major lockdown-related contractions in 2020Q2 and 2021Q3. We also augment the foreign block with the quarterly-average level of the VIX, and order it first, to examine whether explicitly controlling for global financial risk changes the estimated exchange-rate shock. Because the VIX begins in 1990Q1, similar to Manalo et al. (2015), comparisons between models with and without the VIX are made over matched samples so that differences are attributable to the additional variable rather than a change in the

estimation period. Confidence intervals for the impulse responses are constructed using the Kilian (1998) bootstrap-after-bootstrap procedure.

4 Results of Replication

This section first assesses how closely the aggregate results of Manalo et al. (2015) can be reproduced over their original sample, before examining whether the main findings remain stable when more recent observations are incorporated. The discussion focuses on the impulse responses to an identified exchange-rate shock and on forecast error variance decompositions (FEVDs), which provide complementary evidence on the quantitative importance of exchange-rate shocks and the factors driving movements in the Australian dollar.

4.1 Narrow replication

The narrow replication broadly reproduces the aggregate transmission mechanism reported by Manalo et al. (2015). Figure 1 shows the responses to an exchange-rate shock normalised to generate a 10 percent appreciation of the RTWI. Over the original 1985Q1–2013Q2 sample, the appreciation is followed by declines in Australian output, inflation and the cash rate. Real GDP reaches a trough of around 0.31 percent below trend after five quarters, while year-ended trimmed mean inflation falls by around 0.38 percentage points after six quarters. The cash rate subsequently declines by around 0.54 percentage points, reaching its trough after seven quarters. The exchange rate itself gradually returns towards its pre-shock level.

These responses are close to the qualitative and quantitative patterns in the original paper. In particular, the estimated GDP response is almost identical to the contraction of around 0.3 percent reported by Manalo et al. (2015). The inflation and cash-rate responses are somewhat larger in magnitude in our replication, but their direction and timing remain consistent with the original transmission mechanism: an exogenous appreciation reduces domestic activity and inflation and is followed by monetary easing. The differences in magnitude is explainable, given the use of revised historical data and the reconstructed early cash-rate series described in Section 3. Revisions to data can come about for reasons such as more complete source data arrives, annual supply-use benchmarks are incorporated, seasonal factors are re-estimated and occasional historical revisions change methods or sources.

The FEVD results provide a second test of the replication. The *Baseline* column of Table 1 shows that exchange-rate shocks explain only a small share of the forecast error variance of GDP, inflation and the cash rate. For GDP, the contribution remains close to 2 percent or less at the reported horizons. The inflation and cash-rate decompositions differ somewhat more from the published estimates at intermediate horizons, but the overall conclusion is unchanged: autonomous exchange-rate shocks are not an important source of aggregate Australian macroeconomic volatility.

The *Baseline* column of Table 2 also reproduces the central result concerning the determinants of the exchange rate itself. Exchange-rate shocks dominate RTWI variation at very short horizons, accounting for around 87 percent of the one-quarter forecast error variance. Their importance then declines rapidly as the forecast horizon increases. By 40 quarters, exchange-rate shocks account for only around 6 percent of RTWI variation, while foreign shocks account for more than 90 percent. The replication therefore supports the original paper’s interpretation that persistent movements in the Australian real exchange rate are predominantly associated with foreign economic developments rather than exchange-rate shocks in isolation.

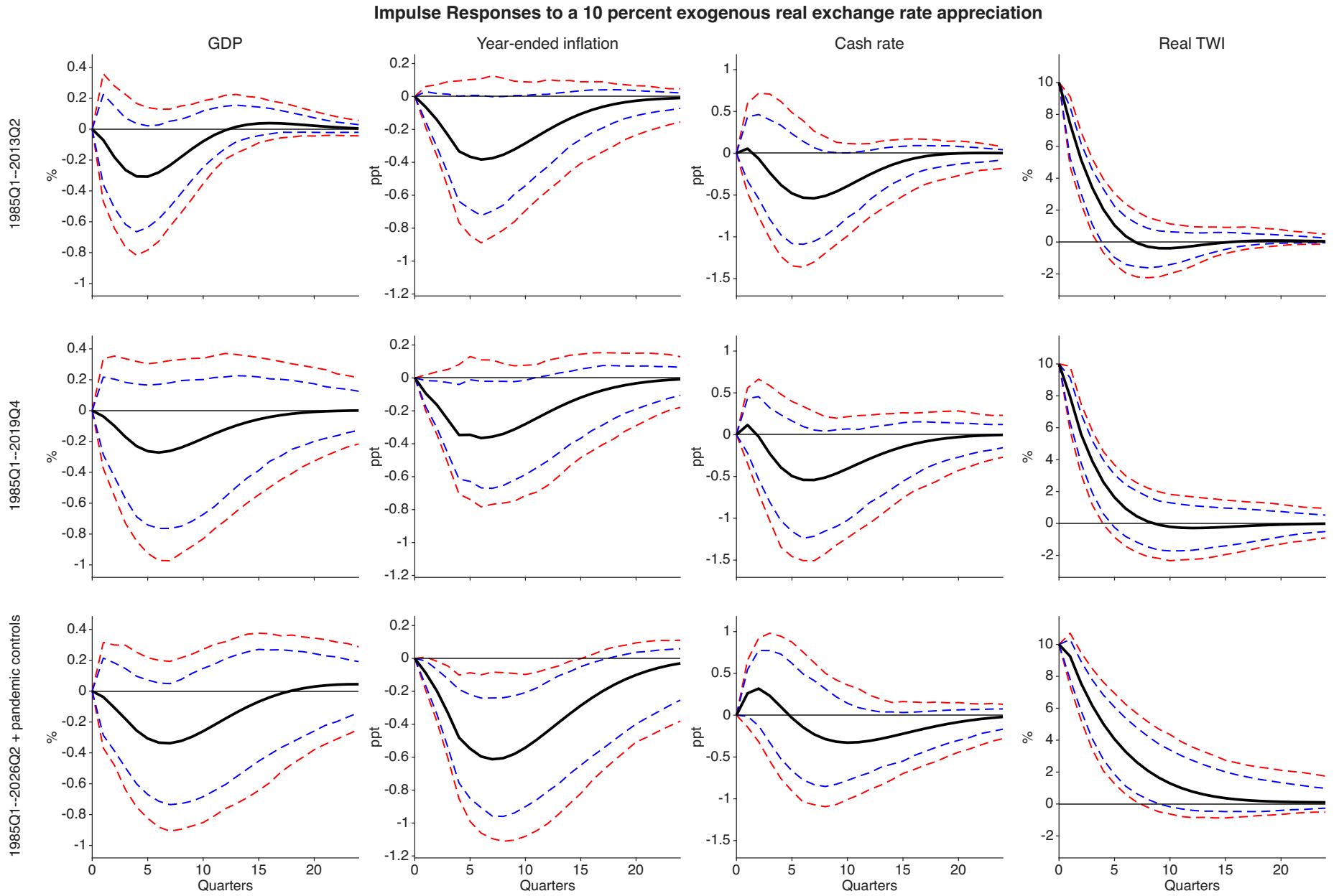


Figure 1: Impulse responses to a 10 percent exogenous appreciation of the real trade-weighted index. Red dashed lines denote 95 percent confidence intervals and blue dashed lines denote 85 percent confidence intervals.

Table 1: Share of forecast error variance of domestic variables explained by exchange-rate shocks, percent.

Variable	Horizon	Manalo et al. (2015)	Baseline	Pre-COVID	Full + controls
GDP	2	0.2	0.1	0.0	0.0
	10	2.6	2.3	1.5	2.8
	20	2.1	1.8	1.5	3.1
	40	1.5	1.4	1.4	3.0
Inflation	2	1.0	0.7	1.8	1.4
	10	3.6	5.7	6.2	15.9
	20	3.9	5.3	6.2	18.2
	40	3.3	4.5	5.8	18.1
Cash rate	2	0.1	0.0	0.1	0.4
	10	1.9	3.4	2.9	1.2
	20	3.8	3.8	3.3	2.4
	40	3.2	3.4	3.1	2.3

Notes: Horizons are measured in quarters. Baseline refers to 1985Q1–2013Q2, Pre-COVID to 1985Q1–2019Q4, and Full + controls to 1985Q1–2026Q2 with pandemic dummies. Published values are from Table A.2 of Manalo et al. (2015).

Table 2: Decomposition of real trade weighted index (RTWI; real exchange rate) forecast error variance by shock group, percent.

Shock group	Horizon	Manalo et al. (2015)	Baseline	Pre-COVID	Full + controls
Foreign	1	11.3	12.1	10.5	6.9
	10	60.1	70.0	58.7	54.2
	20	82.0	85.7	71.5	70.8
	40	92.0	92.4	79.3	78.9
Other Domestic	1	2.6	1.4	0.4	0.3
	10	11.8	8.0	15.5	4.9
	20	5.4	4.0	12.8	3.7
	40	2.4	2.1	9.4	2.8
Exchange Rate	1	86.1	86.5	89.2	92.9
	10	28.1	22.0	25.8	41.0
	20	12.6	10.3	15.6	25.5
	40	5.6	5.5	11.3	18.3

Notes: Horizons are measured in quarters. Foreign shocks comprise shocks to US real GDP and the terms of trade. Other domestic comprises Australian GDP, inflation and cash-rate shocks; Exchange rate denotes the identified RTWI shock. Baseline refers to 1985Q1–2013Q2, Pre-COVID to 1985Q1–2019Q4, and Full + controls to 1985Q1–2026Q2 with pandemic dummies. Published values are from Table A.4 of Manalo et al. (2015). Columns may not sum exactly to 100 because of rounding.

4.2 Stability across samples and the pandemic

The second and third rows of Figure 1 examine whether this transmission mechanism remains stable as the estimation sample is extended. Extending the sample endpoint from 2013Q2 to 2019Q4 produces only modest changes in the main impulse responses. The GDP response remains contractionary, with a trough of around 0.27 percent, while the inflation and cash-rate responses remain close to those obtained in the baseline model.

The timing of the responses is also very similar. The pre-COVID extension therefore provides evidence that the original results are not dependent on the sample ending in 2013.

However, a simple extension of the original specification through 2026Q2 produces a markedly different result. In particular, the GDP response becomes positive following an appreciation rather than displaying the contraction observed in the original and pre-COVID samples. Appendix Figure 2 shows this comparison. Inspection of the reduced-form residuals indicates that the extended sample contains observations that are exceptionally large relative to the historical variation used to estimate the model. The most prominent occurs in 2020Q2, when the standardised residuals for US and Australian real GDP are approximately -9.5 and -8.2 standard deviations respectively. Further unusually large Australian GDP residuals occur during the subsequent lockdown and reopening period.

Since extending a constant-parameter VAR through the pandemic can materially alter the estimated dynamics, the preferred full-sample specification includes controls for the major contractionary quarters of 2020Q2 and 2021Q3. Appendix Figure 3 shows that introducing these controls restores the contractionary GDP response. Now, in the preferred 1985Q1–2026Q2 specification, we find GDP reaches a trough of around 0.34 percent after seven quarters, similar to the baseline point estimate. Nonetheless, the response of year-ended inflation is larger than in the earlier samples, falling by around 0.61 percentage points, while the cash-rate response is more muted, with a trough of around 0.33 percentage points. Hence, once the unusual pandemic-related observations are accommodated for via deterministic controls, the direction and broad sequencing of the original transmission mechanism re-emerged.

Though the point estimates therefore remain economically consistent with the original mechanism, there is considerable uncertainty around some of these responses. In the pandemic-controlled full sample, the 95 percent confidence interval for GDP includes zero at all horizons, as does the corresponding interval for the cash rate. In contrast, the decline in inflation is statistically significant at the 95 percent level from approximately two to fifteen quarters after the shock, compared to the statistically insignificant (at 95% confidence) inflation impulse-response point estimates in earlier samples (both original paper sample and pre-COVID). The stronger inflation response in the post-2019 sample may be consistent with state-dependent exchange-rate pass-through. Carrière-Swallow et al. (2025) find that pass-through to domestic prices is significantly larger when inflation is high and uncertainty is elevated. Recent RBA analysis similarly suggests that external cost pressures may pass through more quickly when inflation expectations and capacity pressures are elevated (Reserve Bank of Australia 2026). While the present SVAR does not identify this mechanism directly, it offers a plausible explanation for the stronger exchange-rate–inflation transmission.

The FEVDs show that the effects of extending the sample are more pronounced for some variables. Table 1 indicates that exchange-rate shocks continue to explain very little of GDP forecast error variance across all three estimation samples. The cash-rate contribution also remains modest. For inflation, however, the contribution of exchange-rate shocks increases substantially in the pandemic-controlled full sample, accounting for around 16–18 percent of the forecast error variance at horizons between 10 and 40 quarters. This is considerably larger than in both the original paper and the earlier replication samples. The conclusion that exchange-rate shocks explain little aggregate output volatility is therefore particularly robust, whereas their importance for inflation appears to have increased in the more recent sample.

A qualification also emerges from the decomposition of the RTWI in Table 2. In every specification, exchange-rate shocks dominate at short horizons and foreign shocks become progressively more important as the horizon lengthens. However, the transition towards foreign-shock dominance is noticeably slower in the extended samples (both pre-COVID and post-2019 periods). In the baseline replication, foreign shocks account for around 92 percent of RTWI forecast error variance after 40 quarters and exchange-rate shocks for only around 6 percent. In the pre-COVID specification, these shares are approximately 79 and 11 percent respectively, while in the pandemic-controlled full sample, they are around 79 and 18 percent. At the 10-quarter horizon, the full-sample specification attributes around 41 percent of RTWI variation to exchange-rate shocks, compared with around 22 percent in the baseline.

The wider replication therefore provides support for the original paper’s shock-absorber interpretation, with some qualifications. Foreign shocks remain the dominant source of longer-horizon movements in the Australian real exchange rate, but their dominance is less strong in the updated samples. Correspondingly, a larger share of medium- and long-horizon exchange-rate variation is attributed to the exchange-rate shock itself. The main domestic transmission mechanism remains broadly intact, but the exchange-rate–inflation transmission is stronger and the variance decomposition suggests that the relative importance of the shocks underlying exchange-rate movements has changed over time.

4.2.1 Global financial risk extension

As a final extension, the foreign block is augmented with the VIX to examine whether some movements identified as autonomous exchange-rate shocks instead reflect changes in global financial risk. The comparison is made using matched samples because the VIX is only available from 1990Q1. The impulse responses remain broadly consistent with the baseline transmission mechanism, suggesting that explicitly controlling for global risk does not overturn the estimated effect of an appreciation on domestic activity and inflation.

The main effect of including the VIX is instead on the attribution of short-run RTWI variation. As shown in Appendix Table 3, adding the VIX reduces the share of RTWI forecast error variance attributed to exchange-rate shocks by around 22 percentage points in the original matched sample, 14 percentage points in the pre-COVID sample and 11 percentage points in the pandemic-controlled full sample. The differences become much smaller at longer horizons, indicating that part of what the baseline SVAR identifies as an exogenous short-run exchange-rate disturbance is associated with global risk conditions that are omitted from the six-variable specification. This interpretation is consistent with Smith (2025), who emphasises the importance of global risk sentiment for Australian financial conditions and short-run exchange-rate movements. At longer horizons, however, explicitly including the VIX has relatively little effect on the share of RTWI variation attributed to exchange-rate shocks. This may leave avenue for future work. A natural extension would be to examine whether richer measures of global financial conditions or alternative identification strategies account for the remaining exchange-rate variation.

5 Conclusion

This paper replicates and extends the aggregate SVAR analysis of Manalo et al. (2015) on the effects of exchange-rate movements in the Australian economy. The narrow replication broadly reproduces the paper’s main findings. A temporary 10 percent exogenous appreciation of the real trade-weighted exchange rate lowers Australian output and inflation and is followed by an easing in the cash rate. Exchange-rate shocks also explain only a small share of aggregate domestic macroeconomic volatility, while foreign shocks become increasingly important in explaining movements in the real exchange rate at longer horizons. Although some quantitative differences remain, particularly for inflation and the cash-rate response, the original transmission mechanism and its main economic interpretation are preserved.

The wider replication shows that these findings are broadly robust to the inclusion of more recent data, but also highlights the importance of how exceptional or unusual observations are treated. Extending the sample to the pre-COVID period leaves the main impulse responses largely unchanged, while simply extending the original specification through the pandemic produces a qualitatively different output response. Residual diagnostics indicate that this result is heavily influenced by the unusually large macroeconomic disturbances associated with COVID-19. Once the major pandemic-related contractions are controlled for, the contractionary output response re-emerges and we observe a stronger and statistically significant exchange-rate–inflation transmission. Importantly, the extended-sample variance decomposition also points to some evolution in the sources of exchange-rate fluctuations: foreign shocks continue to dominate at longer horizons, but exchange-rate shocks account for a larger share of medium- and long-horizon RTWI variation than in the original sample.

The VIX extension provides a further qualification to the interpretation of the identified exchange-rate shock. Explicitly accounting for global financial risk reduces the proportion of short-run RTWI variation attributed to exchange-rate shocks, suggesting that some movements classified as autonomous exchange-rate disturbances in the baseline model are associated with global risk conditions. Overall, the replication provides broad but qualified support for the conclusions of Manalo et al. (2015). The Australian dollar continues to appear to operate primarily as a mechanism through which external disturbances are absorbed, and the domestic effects of an exogenous appreciation remain economically intuitive. At the same time, the updated results show that estimates of exchange-rate transmission and the decomposition of exchange-rate movements are sensitive to major structural disturbances and to the information included in the foreign block of the model.

These results reinforce Manalo et al. (2015) conclusion from a policy perspective that exchange-rate movements should be interpreted jointly with the shocks that generate them, rather than treating an appreciation or depreciation as an independent disturbance. More broadly, however, this exercise shows that robustness of the estimated transmission mechanism need not imply stability in the underlying decomposition of shocks. This is because the impulse responses remain comparatively stable across samples, while the sources attributed to exchange-rate variation changed more materially.

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A Appendix: Additional sample results

A.1 Full sample without pandemic controls

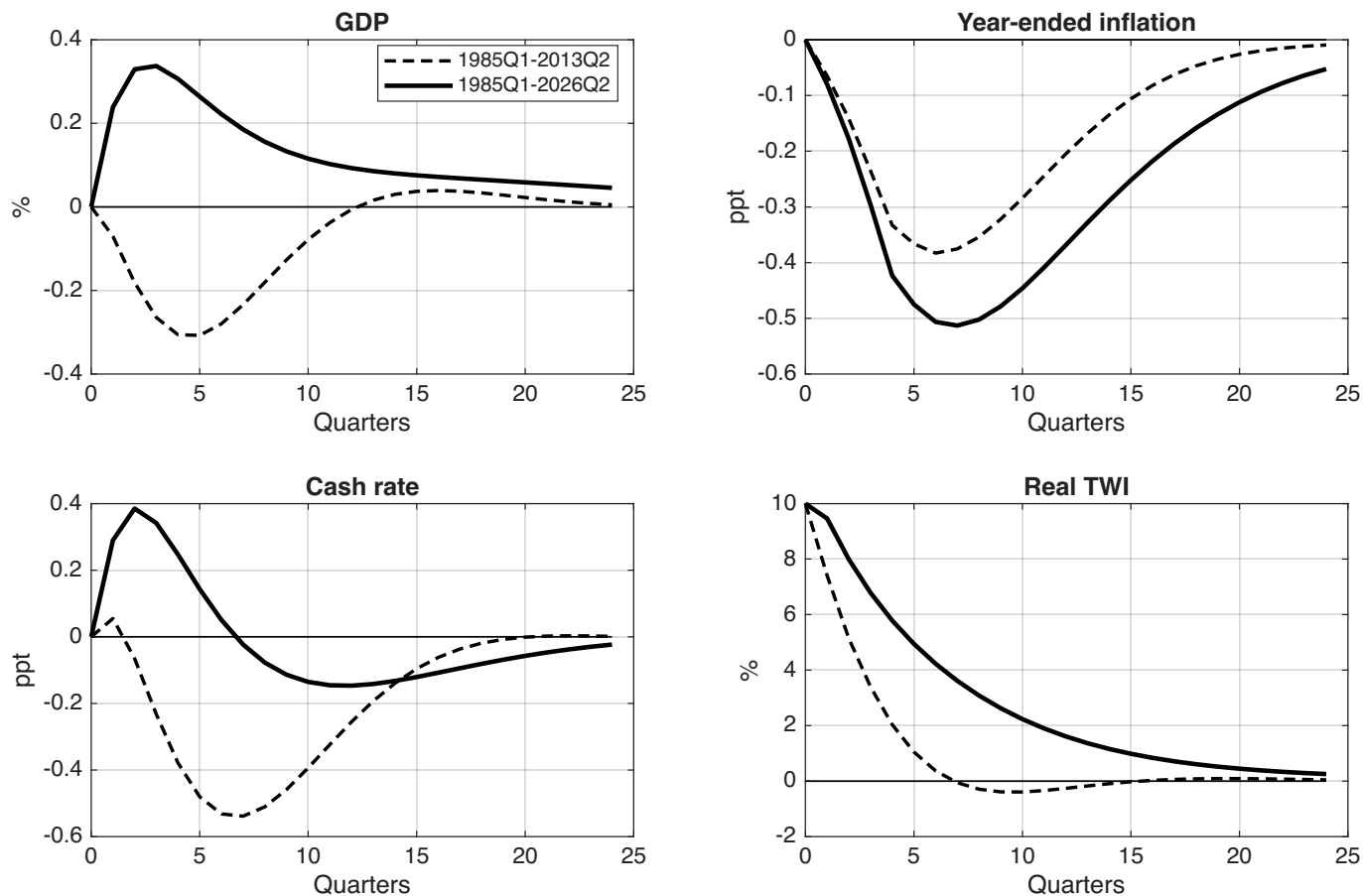


Figure 2: Impulse responses to a 10 percent exogenous appreciation of the real trade-weighted index. Comparison between original sample and full sample without pandemic controls.

A.2 Effect of pandemic controls

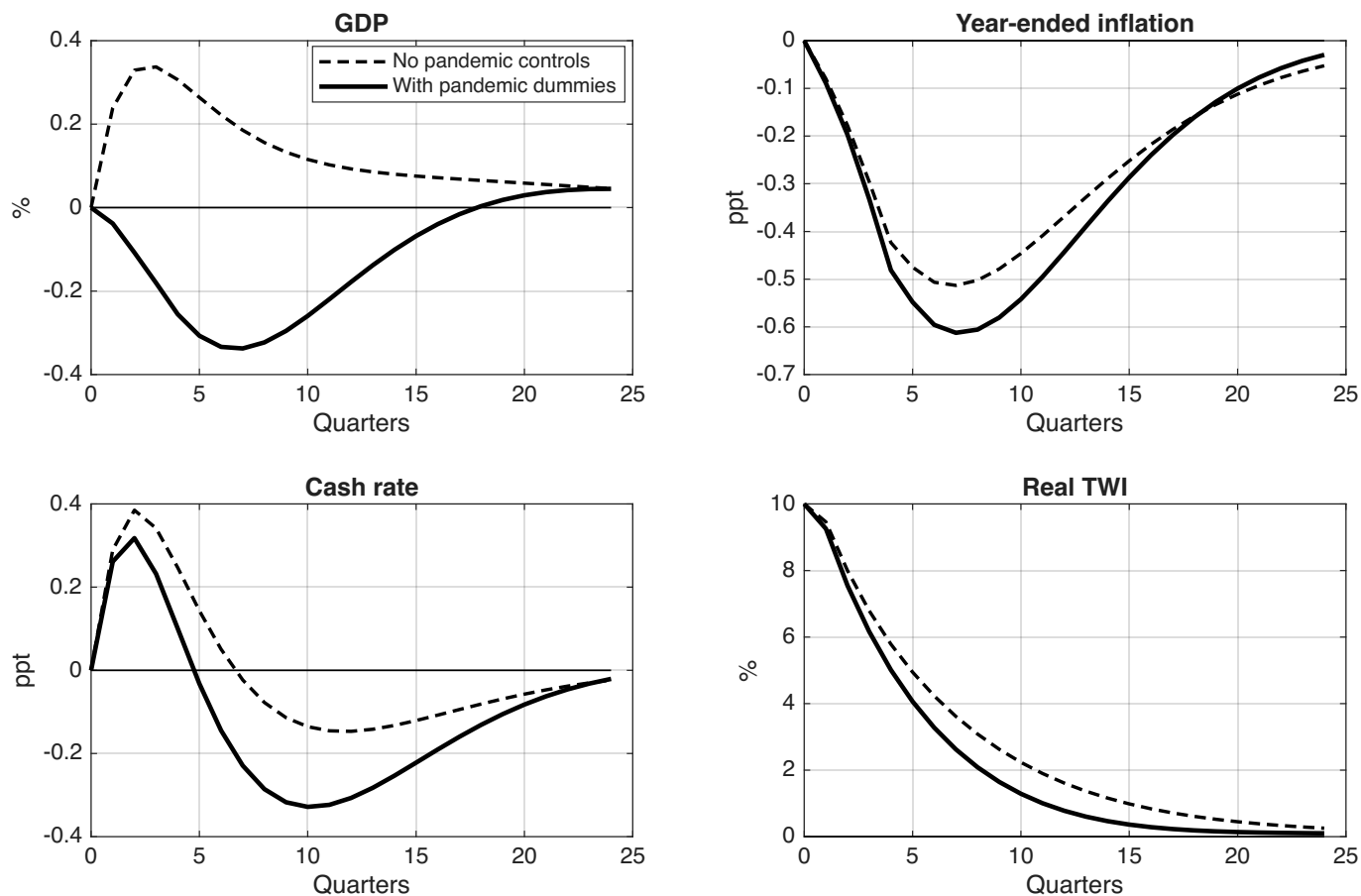


Figure 3: Impulse responses to a 10 percent exogenous appreciation of the real trade-weighted index. Comparison between full sample without pandemic controls and with pandemic controls.

B Appendix: VIX extension

Table 3: Exchange-rate shock contribution to RTWI forecast error variance with and without the VIX, percent.

Sample	Horizon	No VIX	With VIX	Difference
Original sample	1	87.1	64.6	-22.5
	10	25.0	18.6	-6.4
	20	11.8	9.4	-2.4
	40	5.3	4.4	-0.9
Pre-COVID	1	82.5	68.5	-14.0
	10	30.1	28.4	-1.7
	20	15.9	14.8	-1.1
	40	10.4	9.5	-1.0
Full + controls	1	87.8	76.6	-11.2
	10	58.0	58.8	0.8
	20	36.3	37.5	1.3
	40	23.3	22.8	-0.5

Notes: Horizons are measured in quarters. Given data availability of the VIX, Original sample refers to 1990Q1–2013Q2, Pre-COVID to 1990Q1–2019Q4, and Full + controls to 1990Q1–2026Q2 with pandemic dummies. Columns may not sum exactly to 100 because of rounding.

C Acknowledgement of AI Use

I acknowledge using AI (ChatGPT) in three main ways. (1) It assisted in building Matlab code, including identifying where differences were needed from functions provided as part of lecture materials and debugging my code, making graphs, saving results and doing some diagnostics. (2) I discussed my results and my interpretations with it, and it provided helpful suggestions during discussions while I undertook the exercise (e.g., it helped identify where I could go to correctly create my cash rate series based on what other papers did, and provided some options for dummies in extended sample period). (3) It provided some editorial support for my written work, particularly, by making writing more concise and precise, and gave feedback on my final files to catch any oversight or provide suggestions for improvement. Independently, I evaluated the logic, accuracy and value addition of AI code and suggestions, did further research and reading where needed, and adopted suggestions after incorporating my own edits that I deemed necessary.